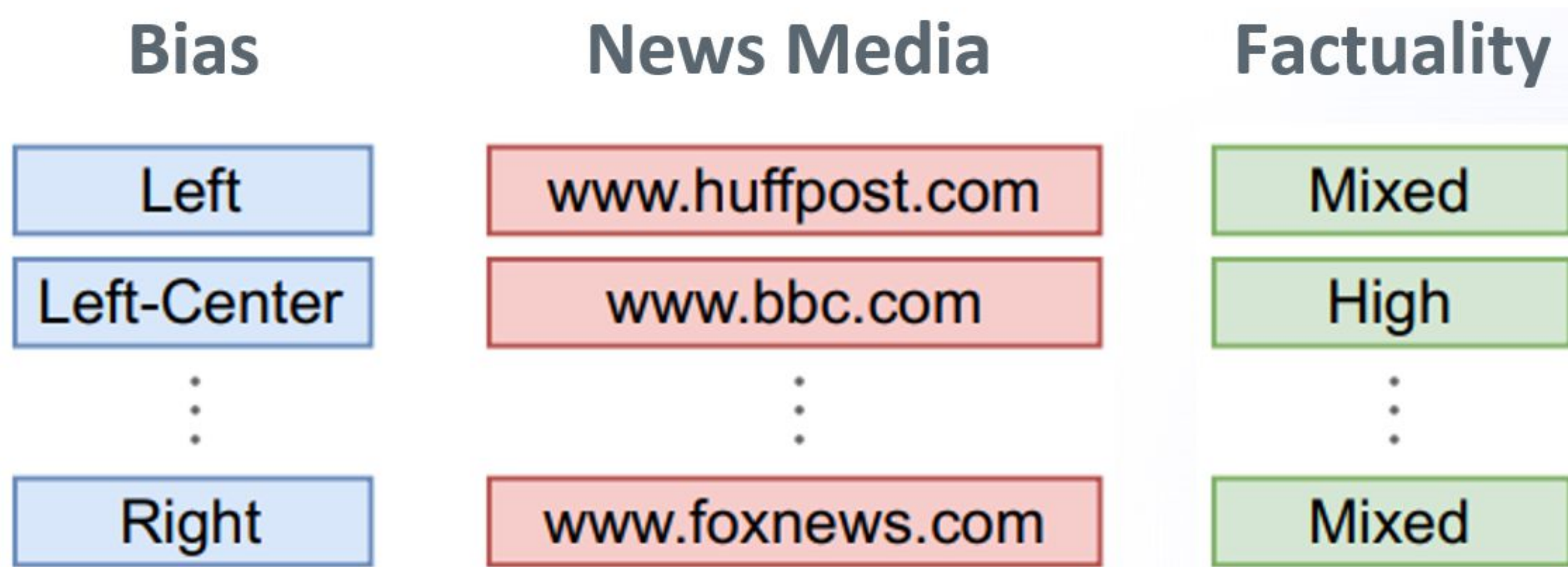


Motivation

News strongly influences people's perceptions, beliefs, and voting behavior, yet there is no standardized way to detect the factuality or political bias of media outlets, nor a comprehensive database of their affiliations and reliability. In today's landscape of rampant mis- and disinformation, it is crucial to empower readers to assess news sources. We propose a novel approach for evaluating entire media outlets by leveraging large language models using handcrafted prompts based on fact-checkers' criteria.

Problem Statement

Given a news outlet's name, predict its bias leaning and factuality level.



Key Contributions

This work proposes a novel methodology that leverages LLMs to predict political **bias** and the **factuality** of the reporting of **entire news media outlets**. Our key contributions are:

- Released a large-scale dataset to model factuality and political bias of news media.
- Emulated the exact criteria used by professional fact-checkers.
- Achieved sizeable improvements over baselines and zero-shot prompting.
- Conducted a comprehensive error analysis on the influence of media popularity and region on model performance.
- Performed an ablation study to evaluate the contribution of dataset components.

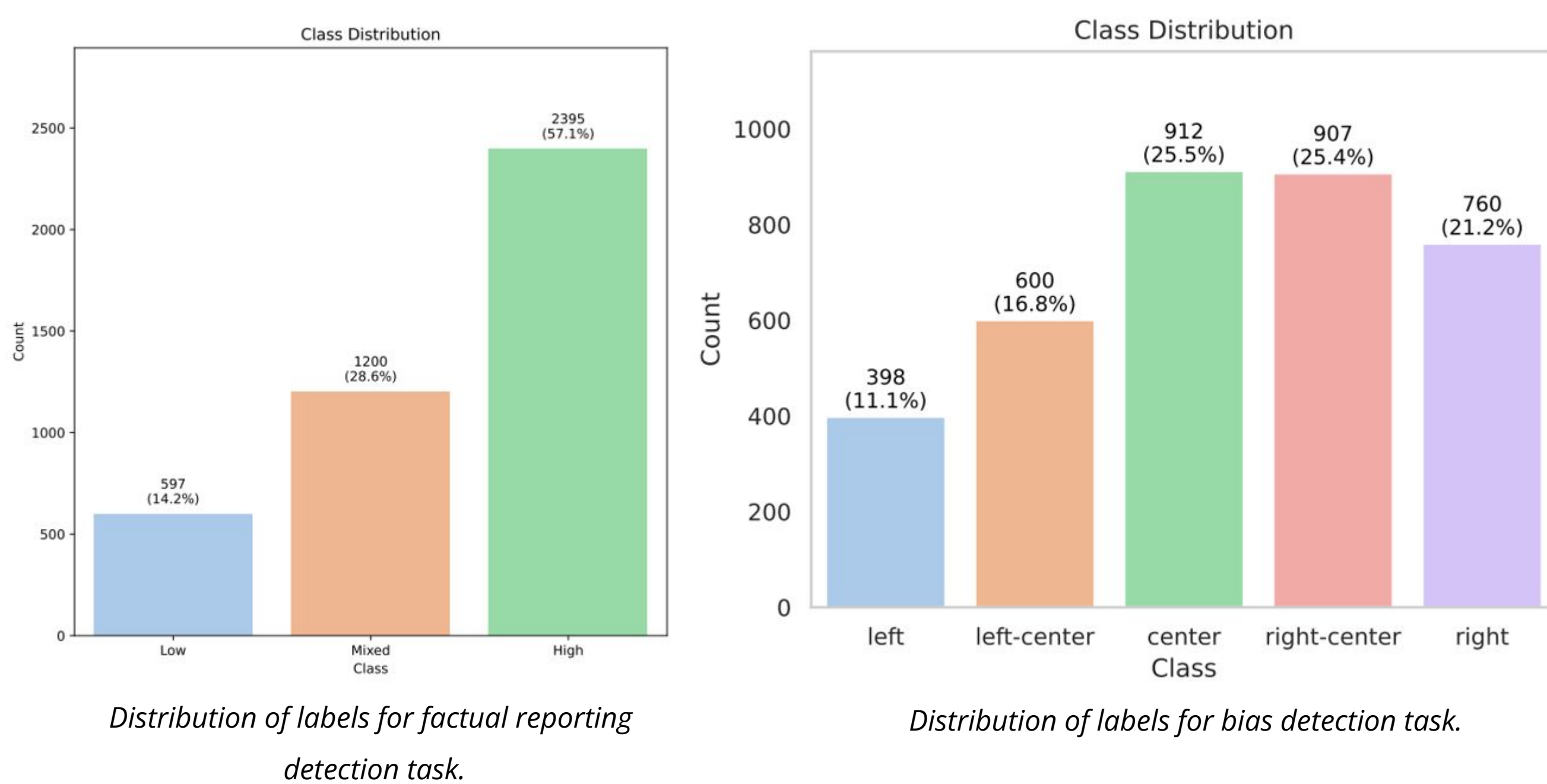
Dataset



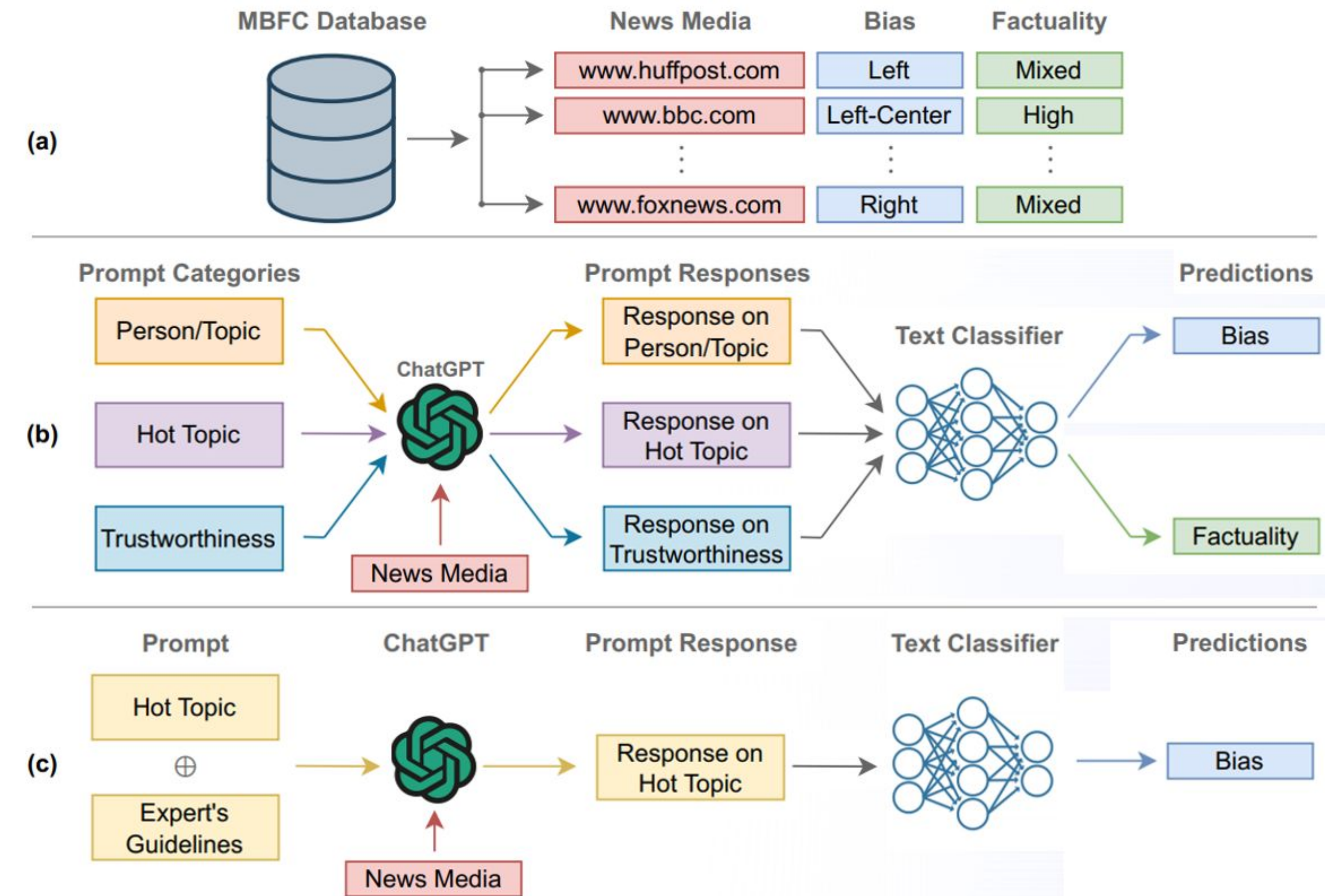
Factual Reporting
Very High
High
Mostly Factual
MIXED
Low
Very Low

Fox News (foxnews.com) – Bias and Credibility

Labels are sourced from Media Bias/Fact Check, covering thousands of English-language outlets with both factuality and bias annotations.



Methodology



Our methodology consists of **two** main prompting strategies:

- Uses handcrafted prompts to query the LLM on public figures, hot topics, and source credibility.
- Systematically embeds expert guidelines into the prompts.

We also conduct experiments in the **zero-shot** setting, where we only provide the media name—or include a part of its article excerpt—to predict political bias and factuality.

Experiments and Results

Class →	Low			Mixed			High			Acc.↑	MAE↓
Model ↓	Pre.	Rec.	F1	Pre.	Rec.	F1	Pre.	Rec.	F1		
Majority Class Baseline											
Majority class	.000	.000	.000	.000	.000	.000	.571	1.000	.727	.571	.572
Zero-Shot Baselines: LLM Prompt Given Name of Media											
GPT-3.5 _{Turbo}	.260	.708	.380	.367	.362	.365	.931	.534	.679	.510	.619
Mistral-7B _{Instruct-v0.1}	.181	.233	.204	.288	.678	.404	.750	.187	.300	.335	.753
LLaMA2-7B _{Chat}	.200	.276	.232	.292	.661	.405	.763	.208	.327	.348	.744
LLaMA3-8B _{Chat}	.333	.275	.301	.274	.635	.383	.560	.213	.309	.343	.726
LLaMA3-70B _{Chat}	.473	.792	.592	.352	.471	.403	.839	.555	.668	.565	.471
Zero-Shot Baselines: LLM Prompt Given Articles from Media											
GPT-3.5 _{Turbo}	.508	.882	.645	.812	.394	.531	.600	.455	.517	.580	.610
Mistral-7B _{Instruct-v0.1}	.324	.353	.338	.259	.212	.233	.167	.182	.174	.250	1.040
LLaMA2-7B _{Chat}	.333	.353	.343	.303	.303	.258	.242	.250	.300	.940	.940
LLaMA3-8B _{Chat}	.345	.294	.317	.291	.485	.364	.438	.212	.286	.330	.780
LLaMA3-70B _{Chat}	.705	.912	.795	.586	.515	.548	.741	.606	.667	.680	.360
Our Method (Hand-Crafted Prompts)											
SVM _{TF-IDF}	.736	.650	.690	.685	.671	.678	.878	.912	.895	.806	.206
BERT _{Base}	.629	.650	.639	.683	.575	.624	.858	.919	.887	.782	.238
RoBERTa _{Base}	.658	.642	.650	.676	.608	.640	.874	.923	.897	.793	.219
DistilBERT _{Base}	.672	.650	.661	.668	.629	.648	.875	.908	.891	.791	.222

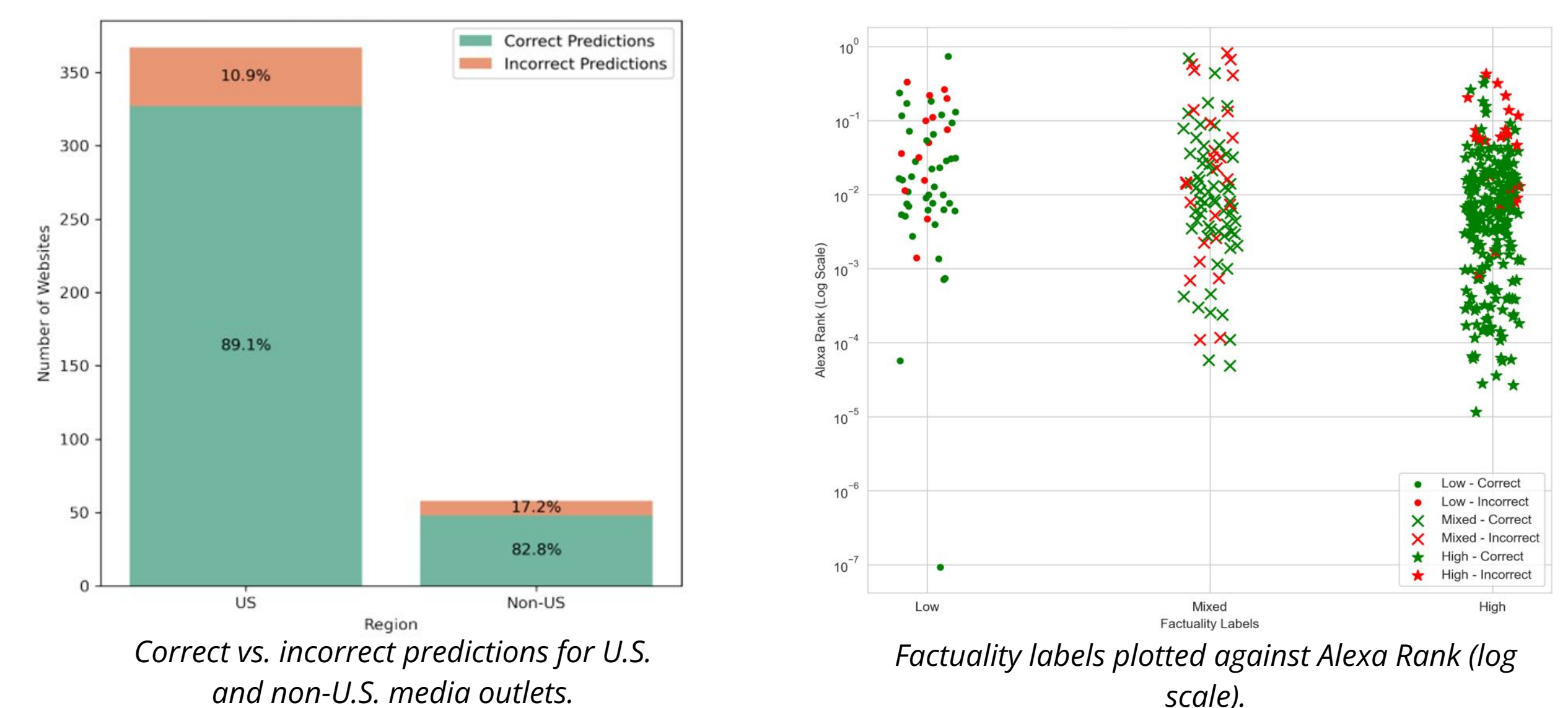
Table 1: Results for *factuality* prediction. **Bold** values indicate the best scores for each category.

Class →	Left			Center			Right			Acc.↑	MAE↓
Model ↓	Pre. Rec. F1	Pre. Rec. F1	Pre. Rec. F1	Pre. Rec. F1	Pre. Rec. F1	Pre. Rec. F1	Pre. Rec. F1	Pre. Rec. F1			
Majority Class Baseline											
Majority class	.000 .000 .000	.427 1.000 .598	.000 .000 .000	.427 1.000 .598	.000 .000 .000	.427 1.000 .598	.000 .000 .000	.427 1.000 .598	.000 .000 .000		
Zero-Shot Baselines: LLM Prompt Given Name of Media											
GPT-3.5 _{Turbo}	.398	.537	.457	.744	.699	.721	.685	.614	.648	.636	.497
Mistral-7B _{Instruct-v0.1}	.882	.750	.811	.824	.945	.880	.927	.843	.883	.869	.152
LLaMA2-7B _{Chat}	.870	.750	.805	.823	.940	.878	.927	.843	.883	.867	.154
LLaMA3-8B _{Chat}	.318	.762	.449	.714	.192	.303	.727	.819	.771	.542	.540
LLaMA3-70B _{Chat}	.855	.738	.792	.877	.934	.905	.878	.873	.875	.874	.168
Zero-Shot Baselines: LLM Prompt Given Articles from Media											
GPT-3.5 _{Turbo}	.596	.848	.700	1.000	.647	.786	.742	.697	.719	.730	.420
Mistral-7B _{Instruct-v0.1}	.1310	.273	.290	.352	.576	.437	.235	.118	.157	.320	.870
LLaMA2-7B _{Chat}	.421	.485	.451	.391	.545	.456	.500	.235	.320	.420	.730
LLaMA3-8B _{Chat}	.290	.545	.379	.556	.147	.233	.448	.394	.419	.360	.950
LLaMA3-70B _{Chat}	.722	.788	.754	.800	.706	.750	.735	.758	.746	.750	.340
Our Method (Hand-Crafted-Systematic Prompts)											
SVM _{TF-IDF}	.914	.800	.853	.915	.940	.927	.883	.910	.896	.902	.133
SVM _{TF-IDF}	1.000	.850	.919	.859	.962	.907	.942	.886	.913	.911	.093
BERT _{Base}	.859	.762	.808	.887	.902	.894	.884	.916	.899	.881	.147
BERT _{Base}	.949	.925	.937	.908	.967	.937	.962	.904	.932	.935	.075
RoBERTa _{Base}	.827	.775	.800	.918	.913	.915	.901	.934	.917	.895	.138
RoBERTa _{Base}	.923	.900	.911	.877	.934	.905	.936	.880	.907	.907	.103
DistilBERT _{Base}	.797	.787	.792	.885	.885	.885	.892	.898	.895	.872	.159
DistilBERT _{Base}	.912	.912	.912	.862	.923	.892	.928	.855	.890	.895	.114

Table 2: Results for *political bias* prediction (3-point scale). Each model marked with the † symbol indicates that it is trained on data derived from prompts incorporating expert guidelines. **Bold** values indicate the best scores for each category.

For **factuality prediction**, our prompt-based methods achieve **~80%** accuracy, outperforming prior work, with SVM + TF-IDF beating fine-tuned transformers. Zero-shot LLMs perform poorly with media names alone, but adding articles boosts accuracy by **~12%**. For **political bias prediction**, systematic prompts raise accuracy to **~93%**, outperforming handcrafted prompts and zero-shot LLMs, with LLaMA3-70B leading zero-shot results but still trailing our approach.

Impact of Media Popularity



Accuracy improves with outlet popularity, with popular sites showing better performance while accuracy drops for less popular outlets. US-based outlets also achieve higher accuracy, highlighting geographic bias in LLM data.

Conclusion and Future Work

We used LLMs with professional fact-checker criteria to predict news media factuality and political bias. Our error analysis showed media popularity affects performance, and combining handcrafted with systematic prompts gave the most accurate results.

Our future work will cover the following:

- Refine factuality prompts with embedded expert methodologies.
- Expand political bias detection beyond U.S.-centric labels.
- Jointly predict factuality and political bias for comprehensive assessment.
- Explore prompt learning and optimization techniques.
- Use open-source, instruction-tuned models for better reproducibility.